

Fertility, Housing, and Population Dynamics

From a stationary closure to a dated transition

Tommaso De Santo

August 2026

Today: from the population closure to an equilibrium transition

Last time: we focused on the missing population closure:

births today $\longrightarrow$ new households later $\longrightarrow$ housing demand and prices.

Today: I try to make that idea an equilibrium object:

- build the simplest model with a well-defined steady state;
- separate the immediate response to a fertility-preference shock from the later demographic adjustment;
- add the same population accounting to the existing quantitative model;
- ask whether the 2007–2023 economy can be interpreted as a point along such a transition.

The update is small: the same household architecture, a population closure, and one nested first-child housing term.

A toy economy

To fix ideas, let's look at a very simplified economy with young and old households:

- Y_t is the number of young households at date t ; O_t is the number of old households.
- Total adult-household population is $S_t = Y_t + O_t$.
- A young household chooses consumption, housing services, and n_t **household children**. Next period it becomes old.
- In this unitary-household model, each household child becomes one young household in the next generation.
- Both generations purchase housing services in the same market at price p_t .

The cohort law is therefore

$$Y_{t+1} = n_t Y_t, \quad O_{t+1} = Y_t,$$

so $n_t = 1$ is replacement.

Focus of the toy model: fertility, cohort replacement, and the housing-services market. The quantitative model adds tenure choice, mortgages, saving, and bequests.

Household choices and the housing market

A young household earns y and chooses consumption c , housing services h , and household children n :

$$u(c, h, n) = c + \alpha \log h + \nu \log n, \quad c + ph + [q + a(p + \omega)]n = y.$$

Here α values adult housing, ν values children, and each household child requires a units of housing plus q other goods. The term ω is the extra cost of accessing child-suitable housing.

The interior choices are

$$n(p) = \frac{\nu}{q + a(p + \omega)}, \quad h_Y(p) = \frac{\alpha}{p} + an(p).$$

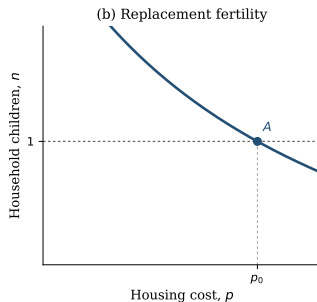
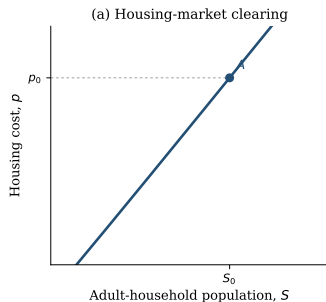
An old household has utility $c_O + \bar{h}_O \log h_O$, so its housing demand is $h_O(p) = \bar{h}_O/p$. Housing supply and market clearing are

$$H^s(p) = \bar{H}p^\varepsilon, \quad Y_t h_Y(p_t) + O_t h_O(p_t) = H^s(p_t).$$

The parameter $\bar{H}$ scales housing supply and ε is its price elasticity.

A higher housing cost reduces both housing demand and fertility. Cohort sizes then feed back into the same market-clearing price.

Stage 1: the initial steady-state equilibrium



A steady state is a constant triple (Y^*, O^*, p^*) . It requires

$$n(p^*) = 1, \quad Y^* = O^*.$$

Replacement selects

$$p_0 = \frac{v_0 - q}{a} - \omega.$$

Housing clearing then selects $S_0 = S^*(p_0)$, where

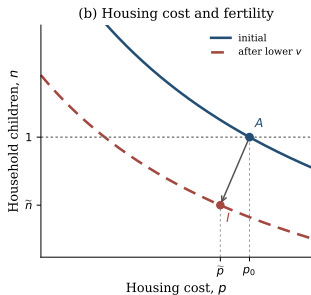
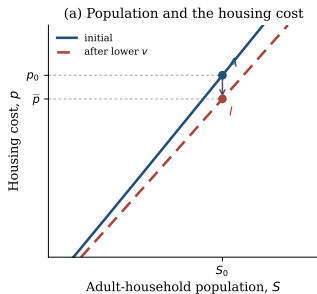
$$S^*(p) = \frac{2\bar{H}p^{1+\varepsilon}}{\alpha + \bar{h}_O + ap}.$$

Along the steady-state schedule, direct differentiation gives

$$\frac{dS^*}{dp} = \frac{2\bar{H}p^\varepsilon [(1 + \varepsilon)(\alpha + \bar{h}_O) + \varepsilon ap]}{(\alpha + \bar{h}_O + ap)^2} > 0 \quad \text{for } \varepsilon \geq 0.$$

Point A is the initial steady state.

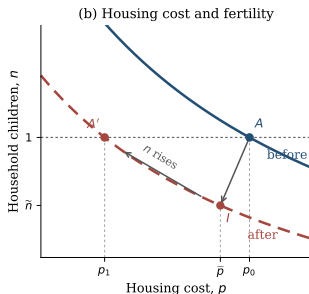
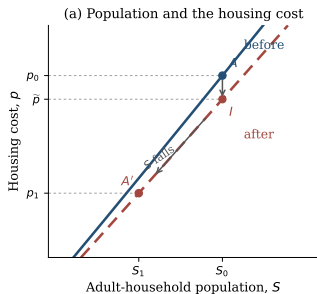
Stage 2: the impact response



- **Shock:** $v_0 \rightarrow v_1 < v_0$.
- **Inherited cohorts:** $Y_t = Y_0$, $O_t = O_0$, $S_t = S_0$.
- **Housing cost:** lower child-related demand gives $p_0 \rightarrow \tilde{p}$.
- **Fertility:** $\tilde{n} = n(\tilde{p}; v_1) < 1$.
- **Point I:** the impact equilibrium with inherited cohort sizes.

Stage 3: demographic adjustment

► Dynamic path



- **Below replacement at l :**

$$Y_{t+1} = \tilde{n}Y_t < Y_t.$$

- **Cohorts:** smaller young cohorts eventually reduce S .
- **Housing cost:** lower demand moves $\tilde{p} \rightarrow p_1$.
- **Fertility:** lower p raises n along the red curve.
- **Point A' :** $n = 1$, $p_1 < p_0$, and

The price response cushions the demographic contraction: falling p raises n , and the larger entrant cohorts slow the subsequent fall in both S and p .

Below-replacement fertility need not lower population immediately

Let $n_t = n(p_t)$ be household children per young household and $\mu_t = O_t/Y_t$ the old-to-young cohort ratio. Population can rise with $n_t < 1$ whenever

$$\mu_t < n_t < 1.$$

A numerical example

$$100 \text{ young} + 70 \text{ old} \longrightarrow 80 \text{ young} + 100 \text{ old}$$

$$n_t = 0.8 < 1, \text{ but total households rise from 170 to 180.}$$

Housing costs can rise as well if the aging cohort retains more housing than is released by the shrinking young cohort:

$$(1 - \mu_t)h_O(p_t) > (1 - n_t)h_Y(p_t).$$

U.S. interpretation: inherited cohorts can keep population and housing costs rising even after fertility falls below replacement.

From the stationary model to a dated transition

Household problem. Households make the existing fertility, tenure, housing, saving, and location choices. The distribution G_t now records household masses as well as characteristics.

Calendar-time accounting.

$$G_{t+1} = \mathcal{T}(G_t, p_t) + E_{t+1}, \quad E_{t+5} = \frac{b_t^{\text{child}}}{2.1}.$$

The existing transition $\mathcal{T}$ moves households across age and economic states. The current run converts literal births into entrant households after twenty years.

Housing market. At each date, the housing cost clears demand from the current distribution:

$$H^s(p_t) = D(G_t, p_t).$$

- **2007–2023:** Census household totals and ACS age shares fix the mass of each age group; the model determines the distribution within each group.
- **After 2023:** the demographic reweighting ends and cohort masses evolve under the model law above.

We calibrate an economy moving from 2007 to 2023

1. **Construct an old steady state.** Choose the initial fertility preference so completed fertility is 2.1. This is a replacement normalization, not a measured 2007 target.
2. **Construct the 2007 state.** Reweight the old distribution only by observed 2007 age shares, preserving wealth, tenure, and fertility states within each age.
3. **Run the economy to 2023.** Feed in Census household totals and ACS age shares. Let the fertility preference decline linearly, and estimate ten household-and-supply parameters and the total preference decline using twelve maintained moments, evaluated mainly in 2023.
4. **Check the path.** Compare untargeted model movements with period fertility, birth timing, ownership, rooms, house prices, and rents.
5. **Continue after 2023.** Hold preferences fixed and let the inherited household distribution and birth pipeline evolve.

The population and age bridge are inputs, not successes of the model. The 2007 economy is a dated initial condition, not literally the old steady state.

2023 calibration

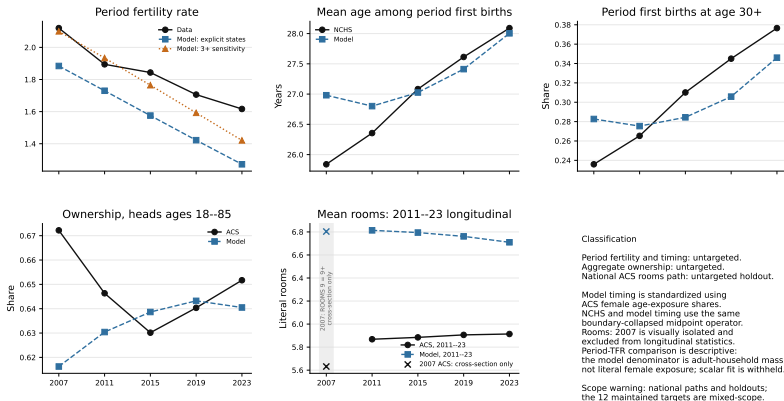
Moment	Target	Model
Completed fertility	1.918	1.918
Childless share	0.188	0.186
Mean age at first birth	26.045	26.327
Share of first births at age 30+	0.260	0.242
Rooms response around first birth	0.720	0.380
Rooms gap: 3+ versus 1–2 children	0.368	0.359
Ownership gap: family versus no family	0.168	0.177
Ownership rate	0.575	0.546
Mean occupied rooms	5.780	6.710
Wealth / annual labor earnings	6.873	6.766
Annual bequests / aggregate wealth	0.0088	0.0086
Old-age wealth p90/p50	3.448	3.396

Main misses: the rooms response around first birth and mean occupied rooms.

The first-birth rooms target remains provisional.

Historical validation: close timing, weak housing fit

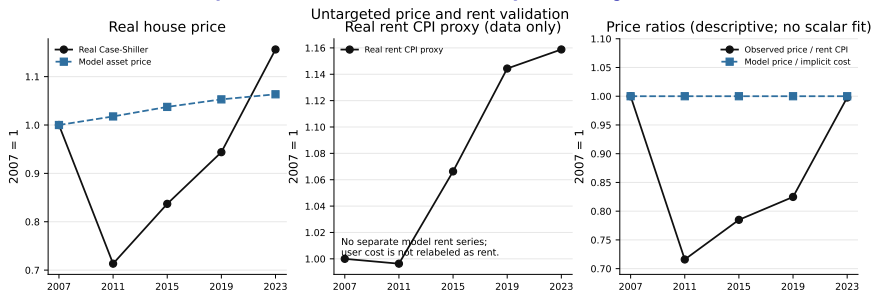
Historical fertility and housing validation



Reading the figure: timing is close by 2023; the model misses the ownership cycle and rooms trend, and its fertility decline is too steep.

National series are holdouts; maintained targets use mixed samples and vintages.

The model does not reproduce the house-price cycle



- The model cannot reproduce the 2007–2011 house-price collapse and later rebound.
- With temporary-equilibrium expectations and a fixed user-cost relation, it has no independent price-to-rent dynamics.

The exercise should therefore be read as a demographic-housing transition, not as a complete account of the post-2007 housing cycle.

Observed house prices: real Case-Shiller. The rent CPI is shown only as a descriptive proxy, not as the model's user cost.

The quantitative options are now explicit

Exercise	Taken from outside the model	What the model determines	Current reading
2007–2023 dated bridge	Household totals and age shares at each date	Prices and household choices within each age group	Useful for fitting 2023; not a demographic validation
Historical semi-open path	Four age-group residual flows	Cohort evolution around those flows	Residuals mainly capture household formation, not immigration
Forward closed path	The fitted 2023 state; no outside entry	Births, cohort sizes, prices, and tenure after 2023	Clean finite-horizon national benchmark; no positive terminal root yet
Forward open path	A fixed outside inflow and retention rate	The same endogenous choices and prices	Sensitivity to continued entry, not a national forecast

For the paper, the clean comparison starts from the fitted 2023 state and shows the closed path together with the open sensitivity.

What the quantitative work has established

- The 2023 model gives a disciplined joint distribution of fertility, tenure, housing, income, and age. Its main remaining misses are housing quantities.
- The current 2007–2023 population and age path is imposed. It provides dated initial conditions, but it does not show that the model generates U.S. demographic history.
- Endogenous fertility first changes the adult-household population after twenty years. The inherited pipeline can therefore raise population and prices temporarily even when current fertility is below replacement.
- With no outside entry, the fitted economy contracts and has no positive steady-state root on the audited price grid. With outside entry, the long-run scale depends on an external closure.

The quantitative result to carry forward is the interaction between fertility, intergenerational housing allocation, and the endogenous housing price—conditional on a stated population closure.

A Coven-style tax reallocates ownership toward families

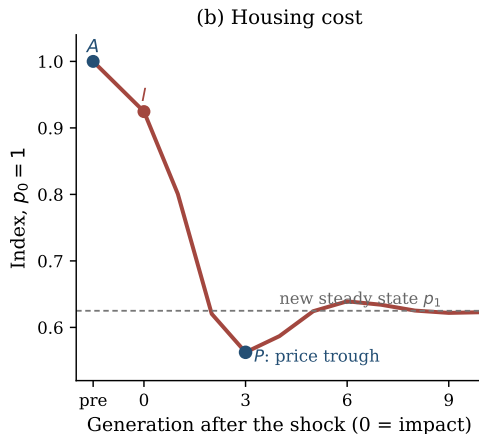
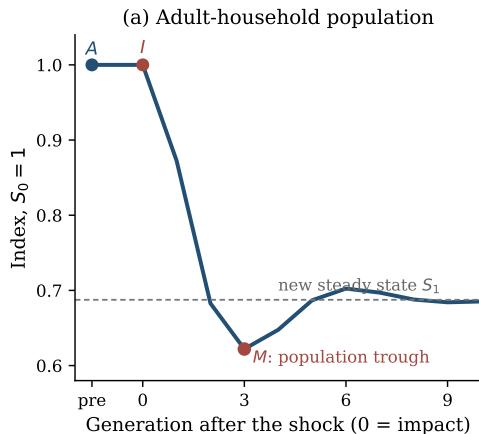
The same fitted 2023 distribution is evaluated under a revenue-neutral property-tax reform. The annual rate rises from 1% to 2%, all revenue is rebated equally, and the house price and transfer clear jointly. Housing supply uses Coven et al.'s California elasticity, $\rho = 0.232$.

	Rebated 1%	Rebated 2%	Change
House price	0.617	0.568	−8.02%
Housing user cost	0.102	0.117	+14.18%
Aggregate ownership	65.15%	66.96%	+1.80 pp
Dependent-child ownership	61.06%	63.59%	+2.53 pp
Births per adult	0.08599	0.08643	+0.52%

Interpretation. Capitalization lowers the upfront purchase price. Ownership rises most for households with dependent children, which are more exposed to down-payment constraints. The fertility response is positive but modest.

Impact mechanism exercise, not a final policy calibration. The supply elasticity is from Coven et al. (2025); the rebated baseline re-anchors the 2023 housing stock by 2.57%.

Population and housing costs can undershoot

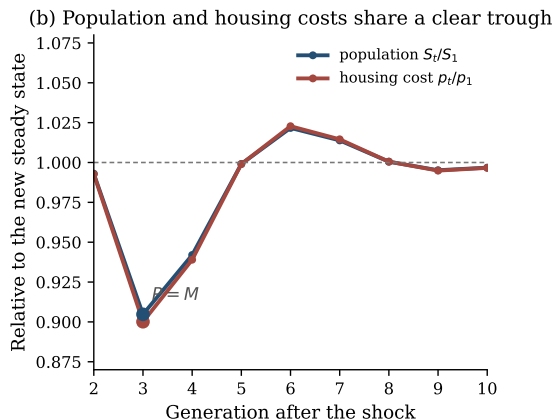
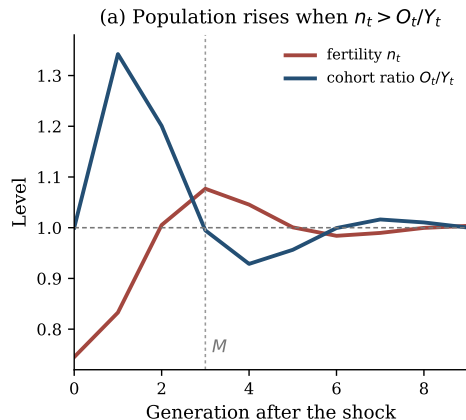


For this parameterization, population and housing costs reach their trough in generation 3 and then converge to the smaller steady state.

Parameters are illustrative; every point solves the model shown in the main slides.

◀ Equilibrium diagram

The cohort ratio determines when population stops falling



Population falls while $n_t < O_t/Y_t$, even after fertility has risen above one. Here the population and price troughs coincide; with different young–old housing demands, they need not.